

Research Sources

Three peer-reviewed studies supporting the Caffeine & Cardiovascular System guide.

SOURCE 01

Energy Drinks and Their Acute Effects on Heart Rhythm in Healthy Children and Teenagers: A Randomized Trial

Authors: Guido Mandilaras, Pengzhu Li, Robert Dalla-Pozza, Nikolaus Haas, Felix Oberhoffer

IDs: PMID 35159306 · PMCID PMC8834195 · DOI 10.3390/cells11030498

Link: [View on PubMed →](#)

WHAT THE STUDY FOUND

- Children and teenagers who drank energy drinks had noticeably more irregular heartbeats than those who did not.
- Surprisingly, their heart rate actually slowed down slightly after drinking. The researchers believe this is because the body tried to bring the heart rate down in response to a sudden rise in blood pressure — a built-in safety reflex.*
- Electrical readings from heart monitors (ECGs) showed changes in how the heart was regulating itself, pointing to stress on the nervous system that controls heart rhythm.*

Conclusion: *Young people with existing heart conditions may be at serious risk of dangerous irregular heartbeats after drinking energy drinks.*

Relevance: *Strong evidence for both blood pressure spikes and irregular heartbeats. The fact that heart rate dropped — while blood pressure spiked — is a useful nuance for your paper: it shows the body actively trying to manage the stress caffeine puts on the heart.*

SOURCE 02

How Red Bull Affects the Heart and Brain Blood Flow in Young Adults: A Randomized Cross-Over Study

Authors: Erik K Grasser, Gayathri Yepuri, Abdul G Dulloo, Jean-Pierre Montani

IDs: PMID 24474552 · PMCID PMC4175045 · DOI 10.1007/s00394-014-0661-8

Link: [View on PubMed →](#)

WHAT THE STUDY FOUND

- Compared to drinking plain water, Red Bull caused a clear rise in blood pressure, heart rate, and the total amount of blood the heart was pumping per minute.
- It increased a measure called the ‘double product’* — essentially a score for how hard the heart is working. A higher score means the heart needs more oxygen to keep up.
- Blood flow to the brain slowed down, while breathing sped up — suggesting the body was under strain beyond just the heart.

Conclusion: *Drinking Red Bull puts the heart under measurably greater strain and reduces healthy blood flow to the brain.*

Relevance: *This is your most direct evidence for Impact 1 (blood pressure) and Impact 2 (heart rate). It uses a well-known, real-world product and tests it against a neutral control, making the findings easy to explain and reference in your paper.*

SOURCE 03

Caffeine and Heart Health: A Full Review of How It Works and What It Does

Authors: Ameer Awashra, Mohammed AbuBaha, et al.

IDs: PMID 41782071 · DOI 10.1186/s40001-026-04089-2

Link: [View on PubMed](#) →

WHAT THE STUDY FOUND

- The relationship between caffeine and heart health is not straightforward. A small or moderate amount of caffeine is often linked to the lowest health risks — it is very high doses that cause the most damage.
- People who drink caffeine regularly build up a tolerance* to it over time. This means their bodies no longer react as strongly to it, so their blood pressure does not spike as sharply as it would in someone who rarely drinks caffeine.
- How your body responds to caffeine depends on your genes, your usual blood pressure, and the type of drink — a can of energy drink hits the body very differently to a cup of coffee.

Conclusion: *Moderate, controlled caffeine intake is generally safe for most people — but the wide variation in how people respond makes it hard to set one rule for everyone.*

Relevance: *Useful for your discussion section. It explains why not every young person reacts the same way to energy drinks, and it supports your alleviation strategies by showing that managing the amount and frequency of caffeine — rather than cutting it out entirely — is a realistic approach.*

Terms Used in This Document

The following terms appear with an asterisk () in the source summaries above. Plain-language definitions are provided here for reference.*

Baroreflex (safety reflex) — The body's automatic system for keeping blood pressure stable. When blood pressure rises too quickly, the brain sends a signal to slow the heart down to compensate.

ECG / Electrocardiogram — A painless test that records the electrical signals in the heart to check how it is beating. Used to detect irregular rhythms or other abnormalities.

QT–RR relationship — A specific measurement from an ECG that shows how the heart's electrical cycle is adjusting to changes in heart rate. Disruptions to this can signal that the heart is under stress.

Double product — A simple calculation (heart rate multiplied by systolic blood pressure) used to estimate how hard the heart is working. The higher the number, the more effort — and oxygen — the heart requires.

Tolerance — When the body gets used to a substance over time and reacts less strongly to it. Someone who drinks caffeine every day will usually feel fewer effects from the same dose than someone who rarely has it.

Irregular heartbeat / Arrhythmia — When the heart beats too fast, too slow, or in an uneven pattern. Some types are harmless; others can be dangerous, especially in people with existing heart conditions.

Supraventricular extrasystoles (SVES) — Extra heartbeats that start in the upper chambers of the heart rather than following the heart's normal rhythm. They often feel like a 'skipped' or 'fluttery' beat.

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